

Demonstrating HydroMod : Constructive Modules for Prototyping Hydraulic Physical Interfaces

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Figure 1: HydroMod. (A) HydroMod supports three types of main modules and four types of additional parts, (B) Snapping each module together by magnets, (C) Actuated seesaw, (D) Silent pin-based color display, (E) Wearable mobile fluid interface.

ABSTRACT

In recent years, actuators that handle fluids such as gases and liquids have been attracting attention for their applications in soft robots and shape-changing interfaces. In the field of HCI, there have been various inflatable prototyping tools that utilize air control, however, very few tools for liquid control have been developed. In this study, we propose HydroMod, novel constructive modules that can easily generate liquid flow and programmatically control liquid flow, with the aim of lowering the barrier to entry for prototyping with liquids. HydroMod consists of palm-sized small modules, which can generate liquid flow with the electrohydrodynamics (EHD) phenomenon by simply connecting the modules. Moreover, users can control the flow path by simply recombining the modules. In this paper, we demonstrate three applications that can be easily prototyped with liquids by recombining HydroMod. These applications showcase the characteristics of using liquids: weight manipulation, color expression, and heating/cooling control.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools.**

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KEYWORDS

Programmable Materials, Toolkit, Prototyping, Fluid Interface, Wearables, Electrohydrodynamics

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1 INTRODUCTION

Groups of components are increasingly being modularized, packaged, and kitted to enable prototyping in electronics and robotics for those without engineering expertise [2, 9, 10, 12, 14, 16]. Toolkits such as littleBits [3], which reduce the time required to connect components and allow users to create electronic systems by simply connecting modules, are also expected to have applications in wearables [7] and soft materials [8]. In the field of HCI, there has been a significant amount of research in recent years on programmable matter, where the material itself has interactivity. Fluid systems, particularly in gases and liquids, are attracting attention for various applications. However, compared with fields such as electronics and robotics, the development of platforms for prototypes in the field of programmable materials that deal with fluid control is a developing area.

Prototyping with materials such as gases and liquids is a challenge because the system that drives the fluid must be built from scratch and requires specialized knowledge. Several toolkits for

handling gases have been released in recent years. Typical examples include The Soft Robotics Toolkit [6], Pneuduino [13] [11], Programmable Air [1], and FlowIO [15]. However, there are still very few toolkits for handling liquids. There are three major reasons why liquids are difficult to handle. First, many large external devices are required for the handling of liquids. Further, relatively large devices such as pumps, tanks, and valves are required to generate a liquid flow. Second, it is difficult to reconfigure the path design of liquid flow. Once the liquid flow path is specified and designed, it takes a considerable amount of time to reconfigure it again using the same parts because the existing fluid system is not designed to be frequently removed. Third, the fluid system is vulnerable. If any part of the system is defective, the working fluid will leak, and the system will not function properly.

In this study, we propose HydroMod, novel constructive modules that can easily generate liquid flow and programmatically control liquid flow, with the aim of lowering the barrier to entry for prototyping through the use of liquids. We utilize electrohydrodynamic (EHD) pumps [5] [4], which use the EHD phenomenon as a pump function to generate liquid flow. The EHD phenomenon occurs when a high voltage is applied to a dielectric liquid. Utilizing the EHD pump, no mechanical moving parts are required because the flow is generated simply by placing an electrode in the liquid and applying a voltage. Furthermore, we expect that the small size and quietness of the EHD pump will address the problem of using a large pump, which is the primary cause of the difficulty when handling a fluid. Therefore, we adopted an EHD pump for the pump function of HydroMod. The HydroMod is composed of palm-sized compact modules: Pump modules that generate liquid flow, Actuate modules that apply a high voltage to drive the Pump modules, and Wire modules that connect the modules. Each module of the HydroMod is designed to be easily connected by magnets. A fluid flow can be generated by simply connecting these modules, and flow paths can be freely designed by simply recombining the modules, which eliminates the difficulty of reconfiguring the fluid system, which is a challenge. Furthermore, because HydroMod is a plug-and-play module, anyone can easily prototype fluid systems without the need for specialized knowledge. In addition, because the fluid system is built through the combination of modules, even if a defect occurs somewhere, it can be fixed by simply changing a part of the module without changing the entire system. Hence, HydroMod is resistant to vulnerabilities in fluid systems.

In this paper, we demonstrate three applications that can be easily prototyped with liquids by recombining HydroMod. These applications showcase the characteristics of using liquids: weight manipulation, color expression, and heating/cooling control.

2 HYDROMOD

2.1 System Overview

HydroMod was created based on the following requirements to enable an easy prototyping with liquid.

- It is possible to design a fluid system that can be easily re-configured by simply combining modules.
- It is an untethered module based on wireless communication, which allows the synchronization of multiple pumps and a distributed placement according to the shape of the target.

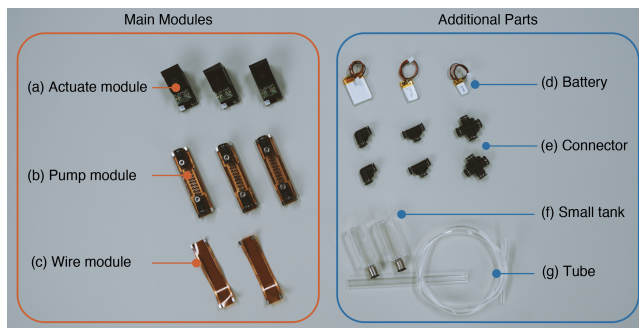


Figure 2: HydroMod supports three types of main modules and four types of additional parts: (a) Actuate module, (b) Pump module, (c) Wire module, (d) Battery, (e) Connector, (f) Small tank, (g) Tube including soft and hard types.

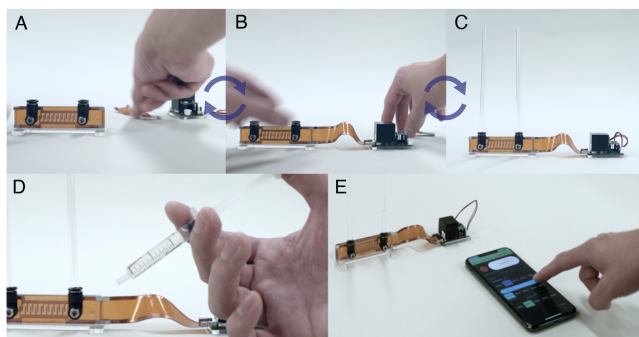


Figure 3: Five step process to operate the fluids with HydroMod: (A) place the Pump module at arbitrary positions, (B) connect the Actuate module to the Pump module with a magnetic connector, (C) attach tubes or small tanks as the flow path, (D) inject Novec 7100 as a liquid in the fluid system, (E) control the created fluid system with simple GUI software.

- The pump function can be miniaturized without the need for large external devices, and fluid systems can be designed including wearable and mobile applications.

As shown in Fig. 2, HydroMod supports three types of main modules and four types of additional parts. The main modules consist of (a) Actuate module that can control the high voltage applied to drive the EHD pump of the Pump module, (b) Pump module with an EHD pump that generates liquid flow as a pump function, and (c) Wire module that acts as a conductor to electrically connect the Pump module and Actuate module. Additional parts consist of (d) batteries that drive the Actuate module, (e) connectors that connect the flow paths, (f) small tanks that store the liquid, and (g) tubes, including soft and hard types, serving as the flow path. The batteries, connectors, and tubes are off-the-shelf components, and the small tanks are prepared by simply crafting acrylic sheets according to the size required.

Each of the modules comprising HydroMod has a built-in magnet as a connector. It is easy to create flow paths and electrical circuits by simply connecting the modules together with the magnets. By reconfiguring the modules, flow paths can be freely designed to take into account the flow direction, pressure, and flow rate. In addition, each module is small enough to fit in the palm of the hand

and has a wireless communication function, making it possible to create a compact hydraulic system that can be carried around and used while moving around, which has been difficult to achieve with existing hydraulic systems.

HydroMod is simple and requires only five steps to operate the fluid system, as shown in Fig. 3. First, the Pump Module, which generates the liquid flow, is placed at an arbitrary position. Subsequently, the Actuate module is connected to the Pump module with a magnetic connector. If the Pump module and Actuate module cannot be connected next to each other owing to placement problems during prototyping, the Wire module can be connected between them to increase the flexibility of the module placement. Next, the additional parts such as connectors, tanks or tubes are attached to create a flow path as needed. The user then injects Novec 7100 as a liquid in the fluid system. Finally, a simple GUI software on a smartphone is used to remotely control the created fluid system. In this way, as the main feature of HydroMod, it allows the users to freely design a fluid system through this simple operation.

2.2 Intuitive Liquid Function Design through the Combination of HydroMod Modules

The main features of the proposed HydroMod are the modularity and reconfigurability of the fluid system. All the devices that make up conventional fluid systems are substituted with small modules, which are easy to handle, and once the modules are connected, they can be easily reconnected. This makes it possible to design the flow path intuitively through trial and error without detailed simulations.

Furthermore, HydroMod has a major feature that is not found in existing fluid systems that handle liquids. In the past, the pressure and flow rates in fluid systems were usually controlled by adjusting the opening of the solenoid valves. However, with HydroMod, the pressure and flow rate can be changed tangibly by reconfiguring the Pump modules. Fig. 4 shows the changes in the fluid functions when the Pump modules are combined in series and parallel. Connecting the Pump modules in series increases the pressure of the fluid, whereas connecting them in parallel increases the flow rate of the fluid. Not only can HydroMod programmatically control the liquid from a microcontroller, but it can also be used to design a fluid system with a tangibly adjusted fluid output in hardware.

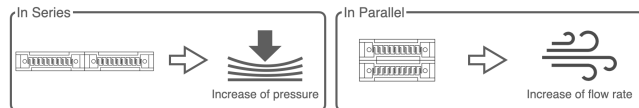


Figure 4: Designing liquid functions by combining pump modules. (Left) If the Pump modules are connected in series, the pressure increases, and (Right) if they are connected in parallel, the flow rate increases.

2.3 Flow Control Method by Recombining HydroMod

The flow direction in HydroMod can be controlled dynamically by changing the position of the Actuate module, without changing the entire fluid system and without introducing additional equipment.

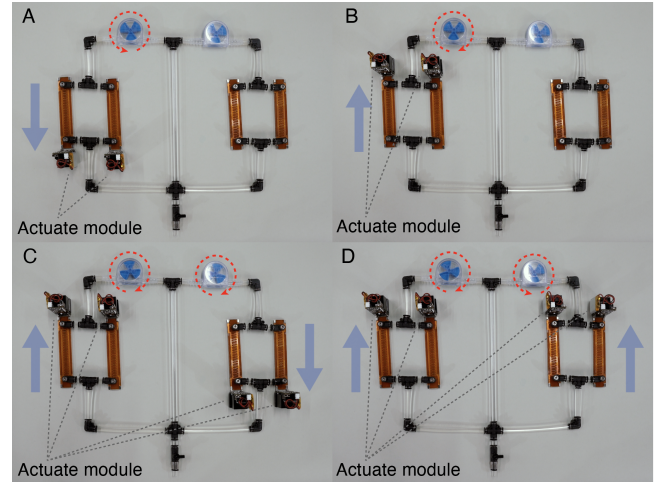


Figure 5: Reconfigurable flow control by recombining Actuate modules: (A) place the Actuate modules in the lower left, only the fan on the left turns counterclockwise. (B) place the Actuate modules in the upper left, only the fan on the left turns clockwise. (C) place the Actuate modules in the upper left and lower right, the fan on the left turns clockwise and the fan on the right turns clockwise simultaneously. (D) place the Actuate modules in the upper left and upper right, the fan on the left turns clockwise and the fan on the right turns counterclockwise simultaneously.

The four consecutive flow controls in Fig.5 are taken as examples and are described in the following order. First, (A) if the two Actuate modules are placed in the lower left, only the fan on the left turns counterclockwise. Subsequently, (B) if the two Actuate modules are placed in the upper left, only the fan on the left turns clockwise. Next, (C) if the two Actuate modules are placed in the upper left and two Actuate modules are placed in the lower right, the fan on the left turns clockwise and the fan on the right turns clockwise simultaneously. Finally, (D) if the two Actuate modules are placed in the upper left and two Actuate modules are placed in the upper right, the fan on the left turns clockwise and the fan on the right turns counterclockwise simultaneously. In this way, fluid systems using HydroMod can dynamically and tangibly control the fluid flow in ways that are not possible with conventional fluid systems. This control method does not require rebuilding the fluid system to change the fluid flow, but can be controlled simply by attaching the Actuate module, and thus there is no need to introduce new equipment.

3 DEMO APPLICATIONS

3.1 Actuated Seesaw

Fig. 6 shows a seesaw object created by combining the modules of HydroMod. By moving the liquid inside the hard tube and tank, the position of the center of gravity is changed, and this application functions as an actuator with kinetic movement. By alternately driving two Actuate modules with different voltage application directions, the liquid moves left and right inside the tube and tank, and the tilt of the seesaw changes as the center of gravity shifts.

In addition, the timing and strength of the pump drive can be controlled remotely through wireless communication, and it is possible to implement with less wiring that does not interfere with the movement of the seesaw. This application is based on the application of weight manipulation, which is one of the properties of liquids.

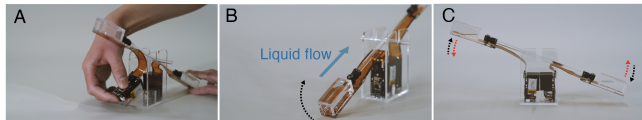


Figure 6: Actuated seesaw. (A) HydroMod allows intuitive design of a fluid system by simply combining modules. (B) It creates a fluid flow and sends it to the face of the seesaw. (C) Kinetic movements are created by moving the liquid inside the tube and tank, and changing the position of the center of gravity.

3.2 Silent Pin-based Color Display

Fig. 7 shows a silent pin-based color display created by combining HydroMod. This pin-based display can silently show a liquid representation synchronized at each drive location, such as a wave representation. The colored water is controlled by utilizing the separation characteristics of the liquid. Novec 7100, which is used as a dielectric liquid, has a high density and is hydrophobic, so it separates from water. When colored water is placed on Novec 7100, Novec 7100 can push the colored water upward, so the colored water can be controlled variably, and it is expressed as a display. This application is based on the application of color expression, which is one of the properties of liquids.



Figure 7: Silent pin-based color display. (A) The liquid in the vertical pin is controlled in an upward and downward direction. (B) By having multiple modules working in synchronization, synchronized liquid expression can be achieved at each driving location. (C) The colored water is controlled by utilizing the separation characteristics of the liquid.

3.3 Wearable Mobile Fluid Interface

Fig. 8 shows a wearable mobile fluid interface created by combining HydroMod. This is a wearable interface that can control cooling by liquid, and can be used for mobile applications, such as moving around or going outside. HydroMod is composed of a combination of palm-sized modules that allows for the portable use of fluid systems, which has been difficult with conventional fluid systems. This application is based on the application of heating/cooling control, which is one of the properties of liquids.

4 CONCLUSION

In this demonstration, with the aim of lowering the barrier to entry for prototyping with liquids, we proposed HydroMod, novel constructive modules that can easily generate and programmatically control liquid flow. HydroMod consists of palm-sized modules, and the flow path can be freely designed by simply recombining the



Figure 8: Wearable mobile fluid interface. (A) A wearable fluid device that can control cooling without external large devices. (B) The pressure can be increased by connecting the Pump modules in series. (C) Cooling the body with wearable mobile fluid interface.

modules. Therefore, HydroMod overcomes the three factors that have made it difficult to handle liquids in the past: the need for large external devices, the difficulty in reconfiguring the flow path design, and the fragility of the fluid system. We presented three applications that can be easily prototyped with liquids by recombining HydroMod. These applications showcase the characteristics of using liquids such as weight manipulation, color expression, and heating/cooling control. We believe that HydroMod will inspire users to explore new types of fluid systems.

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